Sri Lanka Institute of Information Technology



IT3060 - Human-Computer Interaction

Milestone 2

Project Title – **Transport Tracking & Digital Ticketing App**

Assignment Group ID – Y3. S2.WE_62

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Group Member Details & Work Distribution

Name	Student ID	Workload Distribution
Silva D S J	IT23652200	Search, filtering, transport details, route and stop information, and live tracking screens.
Rajapaksha M.P.K	IT23665620	Journey confirmation, passenger details, payment, tickets, history, notifications, cancellation, and profile screens.
Herath H.M.S.G	IT23634626	Staff login, conductor dashboard, journey details, QR scanning, ticket validation, passenger list, incident reporting, and journey completion.
Fernando H L R D	IT23635302	Authority dashboard, live operations, vehicle, route, stop and schedule management, fare and facilities, incident review, and passenger alerts.

1. Project Overview

Ride Track is a mobile application that helps passengers, conductors or ticket inspectors, and transport authority officers manage public bus and train travel through real-time tracking and digital ticketing. User research from Milestone 01 identified problems such as uncertain arrival times, limited live tracking, manual paper tickets, slow ticket validation, and poor visibility of delays and route changes.

The application lets passengers search routes, track vehicles live, and purchase digital QR tickets; conductors scan and validate tickets and report incidents; and authority officers manage vehicles, routes, schedules and passenger alerts through a live dashboard.

Assignment 02 turns these requirements into sketches, wireframes, and low- and high-fidelity prototypes for all three roles, tested with users to identify areas for refinement.

2. Requirement-Mapping

1.1 Main Interface Mapping

ID	Requirement	Affected Interfaces	Role
FR01	Register an account using email or mobile number	Login and registration flow	Shared
FR02	Log in securely	Passenger login; Staff Login; role-based dashboard	Shared
FR03	Search routes and view upcoming arrivals	Passenger Home; Select Transport; Search; Results; Filters	Passenger
FR04	View live location and estimated arrival time	Transport Details; Route and Stops; Live Vehicle Tracking	Passenger
FR05	Purchase a digital QR ticket	Journey Confirmation; Passenger Details; Payment; Booking Confirmation; Digital Ticket	Passenger
FR06	Scan and validate QR tickets	Scan Digital Ticket; Valid and Invalid Results; Validated Passenger List	Conductor or Inspector
FR07	Display vehicle occupancy where available	Search Results; Transport Details; Journey Details; Passenger List	Passenger and Staff
FR08	Notify delays, cancellations, and route changes	Notifications; Report Incident; Incident Review; Passenger Alert	All roles

FR09	Provide live operations dashboard	Authority Dashboard; Live Operations; Incident Management	Authority
FR10	Generate route, delay, and occupancy reports	Authority Dashboard; Live Operations; Incident Management	Authority
FR11	Maintain transactions and trip history	My Tickets; Ticket Details; Journey History	Passenger
FR12	Update profile and preferences	Passenger Profile	Passenger

1.3 Non-Functional Requirement Coverage

NFR	Quality	How the UI addresses it
NFR01	Usability	Consistent navigation, clear labels, progressive flows, and immediate feedback.
NFR02	Performance	Live location and ETA areas are designed for frequent refresh and visible status updates.
NFR03	Security	Role-based entry points, protected payments, and ticket validation states.
NFR04	Availability	Core ticket information remains easy to reach during travel.
NFR05	Privacy	Only necessary passenger and payment fields are requested.
NFR06	Compatibility	Mobile-first passenger and staff screens support contemporary Android layouts.
NFR07	Error handling	Required-field, invalid-ticket, payment-failure, and incident-confirmation states are represented.
NFR08	Accessibility	Readable hierarchy, labelled controls, adequate touch targets, and contrast-conscious design.
NFR09	Scalability	Routes, vehicles, schedules, facilities, and authority records are separated into manageable modules.
NRF10	Offline ticket	The purchased QR ticket is designed as a self-contained scannable view.

2. Sketches Variants

Each member explored the structure of their assigned component before committing to a detailed interface. Sketching focused on information hierarchy, placement of the primary action, navigation continuity, and the visibility of system status. The selected concepts were then translated into consistent wireframes

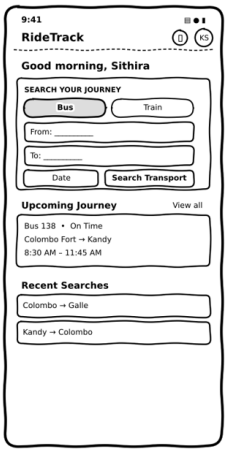
1.3 Variant Selection Rationale

Interface	Selected Direction	Reason
Passenger search and tracking	Search-first home with card-based results and map-centred tracking	Keeps the most frequent task visible and allows users to compare time, price, occupancy, and service status quickly.
Booking and ticketing	Step-by-step confirmation, passenger details, payment, and ticket flow	Reduces cognitive load, validates data at each step, and preserves a clear route back before payment.
Conductor operations	Task dashboard with scan-first validation and explicit result states	Supports rapid ticket checking while clearly distinguishing valid and invalid tickets and recording incidents.
Authority operations	Dashboard and management modules with review-before-publish actions	Makes live disruptions visible and reduces accidental changes to vehicles, routes, schedules, fares, or alerts.

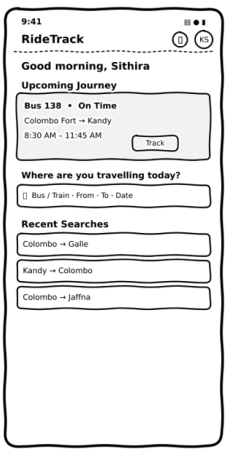
3. Sketches, wireframes & Justifications

3.1.1 passenger Home

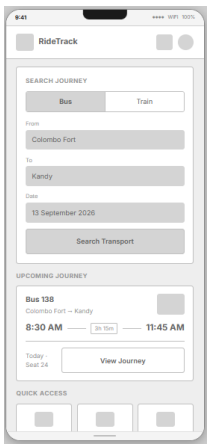
Variante1 (Selected)



Variante2

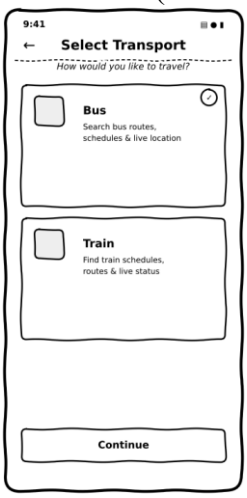


Selected Wireframe

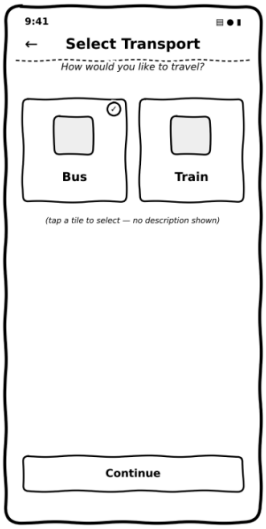


3.1.2 **Select Transport — Bus / Train**

Variant 1(selected)



Variant 2

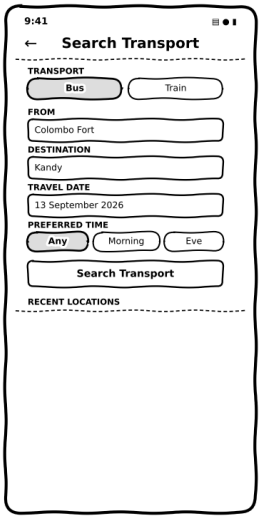


Selected Wireframe

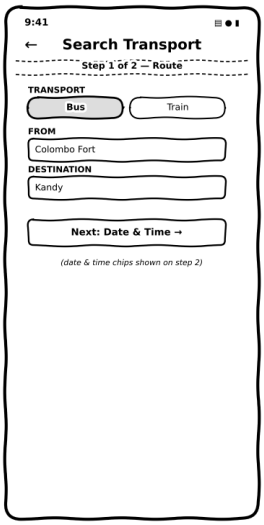


3.1.3 Search Transport

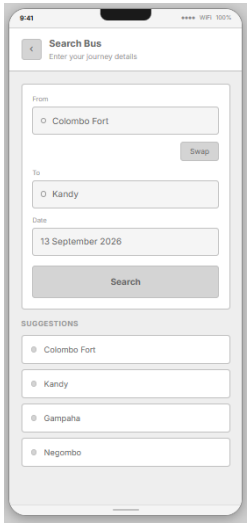
Variant 1(selected)



Variant 2

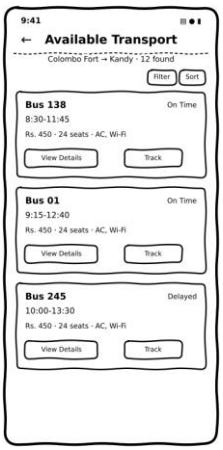


Selected Wireframe



3.1.4 Search Results

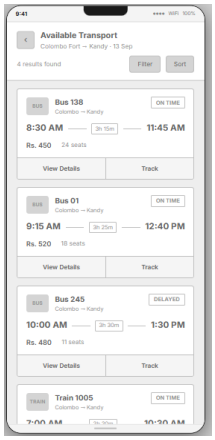
Variant 1(selected)



Variant 2

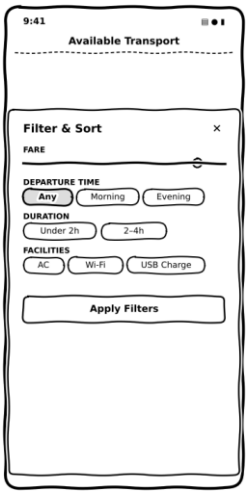


Selected Wireframe

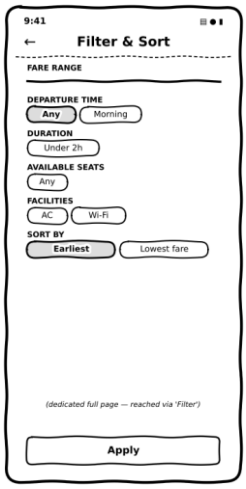


3.1.5 Filter & Sort Results

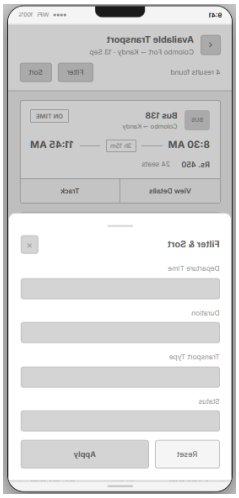
Variant 1(selected)



Variant 2

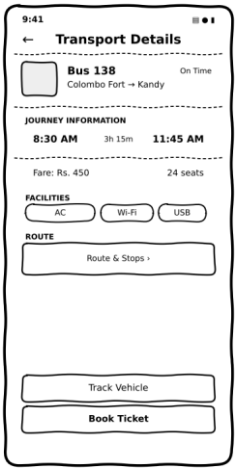


Selected Wireframe

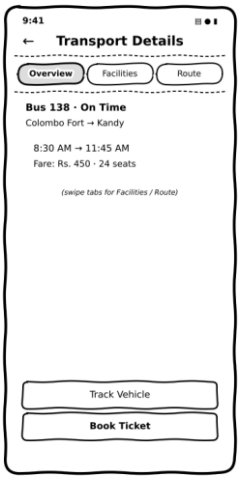


3.1.6 Transport Details

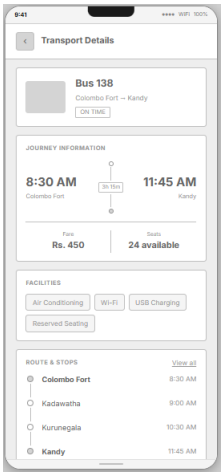
Variant 1(selected)



Variant 2

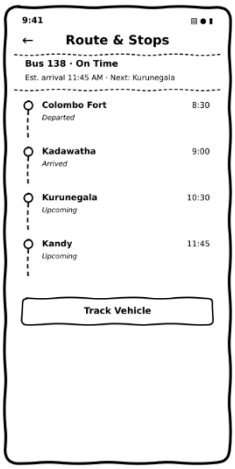


Selected Wireframe



3.1.7 Route and Stops

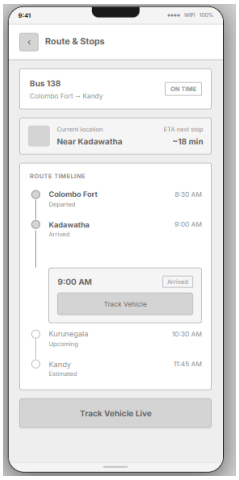
Variant 1(selected)



Variant 2



Selected Wireframe

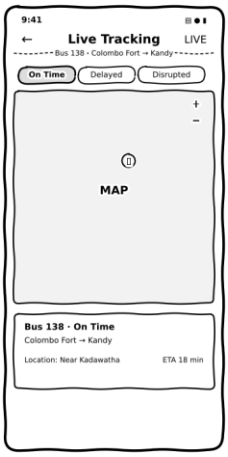


3.1.8 Live Vehicle Tracking

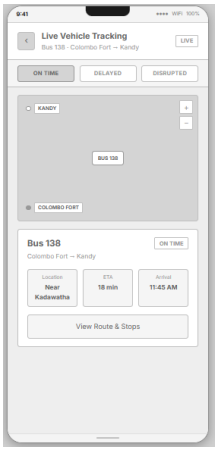
Variant 1 (selected)

Variant 2

Selected Wireframe





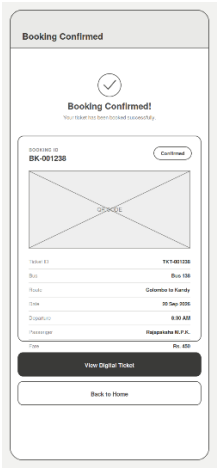


3.1.9 Booking Confirmation

Variant 1(selected) Variant 2 Selected Wireframe





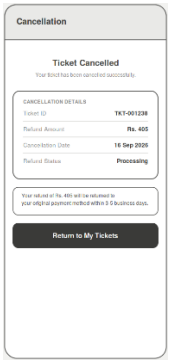


3.1.10 Cancel Booking

Variant 1(selected) Variant 2 Selected Wireframe





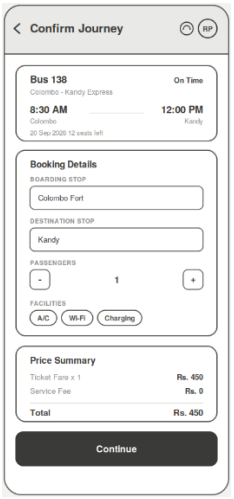


3.1.11 Confirm Journey

Variant 1(selected) Variant 2 Selected Wireframe



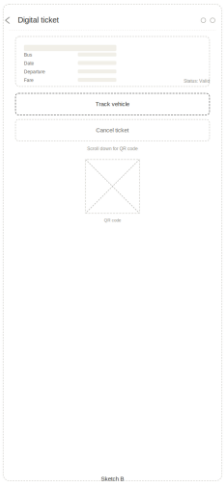


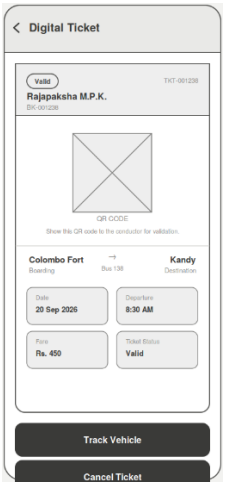


3.1.12 Digital

Variant 1(selected) Variant 2 Selected Wireframe

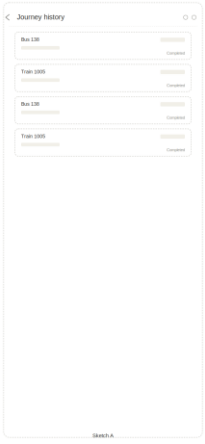


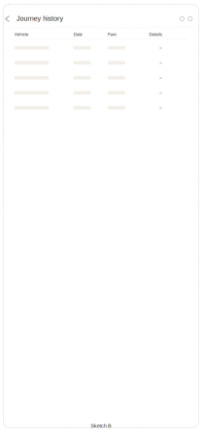


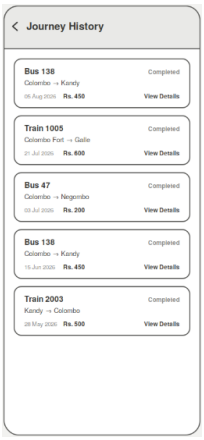


3.1.13 Journey History

Variant 1 Variant 2 (Selected) Selected Wireframe

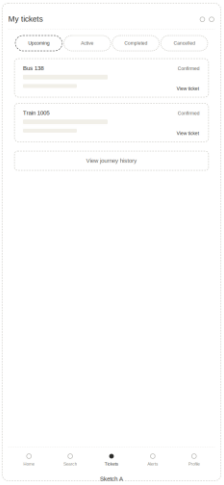


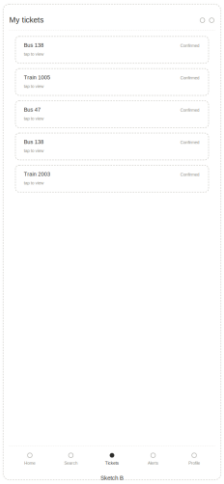


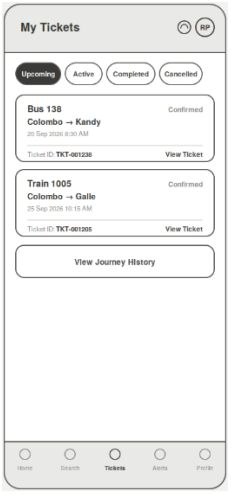


3.1.14 My Tickets

Variant 1(selected) Variant 2 Selected Wireframe

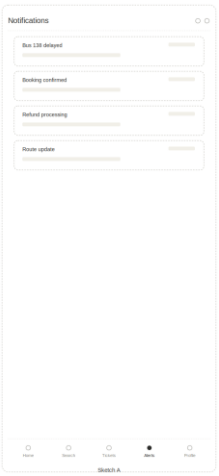




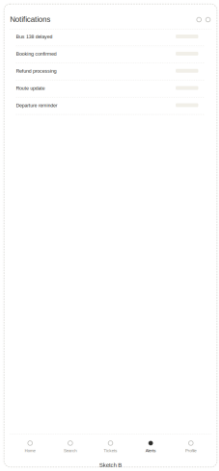


3.1.15 Notifications

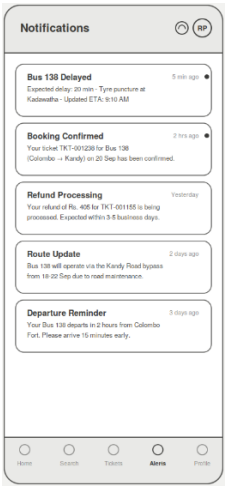
Variant 1(selected)



Variant 2



Selected Wireframe



3.1.16 Passenger Detail

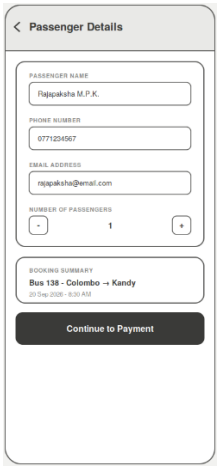
Variant 1(selected)



Variant 2



Selected Wireframe

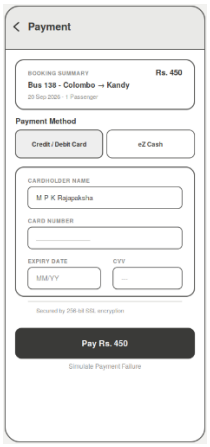


3.1.17 Payment

Variant 1(selected) Variant 2 Selected Wireframe





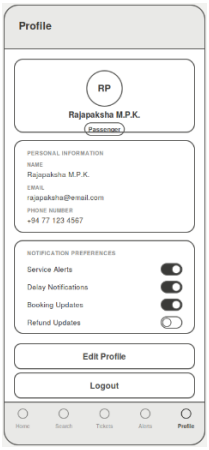


3.1.18 Profile

Variant 1(selected) Variant 2 Selected Wireframe





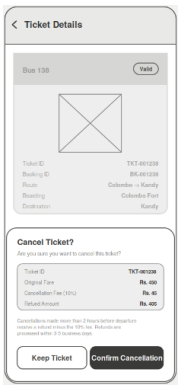


3.1.19 Ticket Details

Variant 1(selected) Variant 2 Selected Wireframe

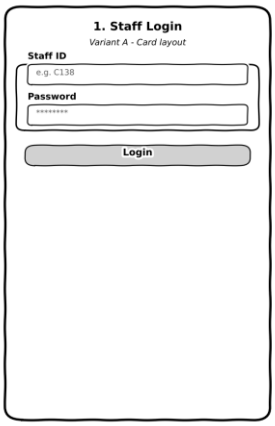




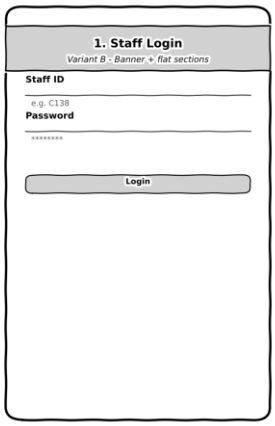


3.1.20 Staff Login

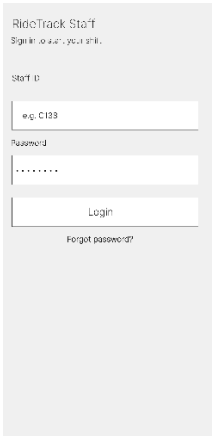
Variant 1



Variant 2

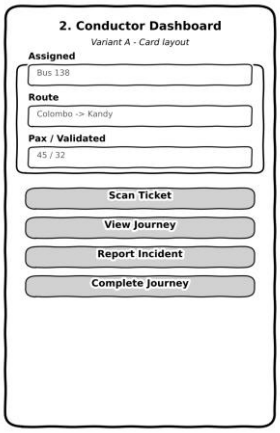


Selected Wireframe

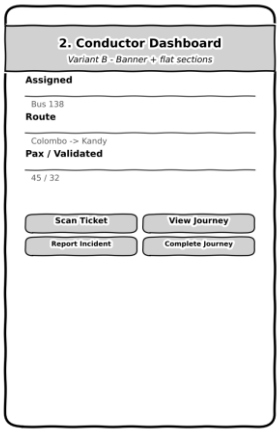


3.1.21 Conductor Dashboard

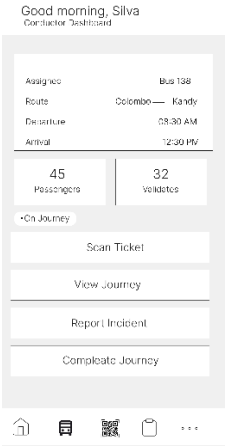
Variant 1(selected)



Variant 2

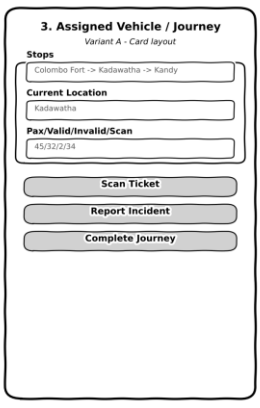


Selected Wireframe



3.1.22 Assigned Vehicle/Journey

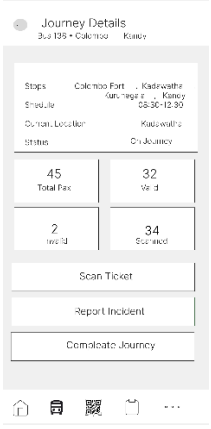
Variant 1(selected)



Variant 2

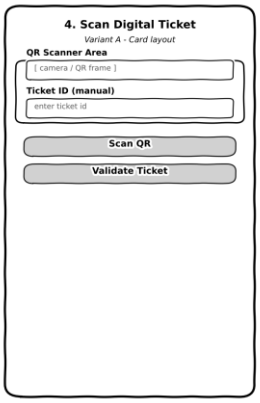


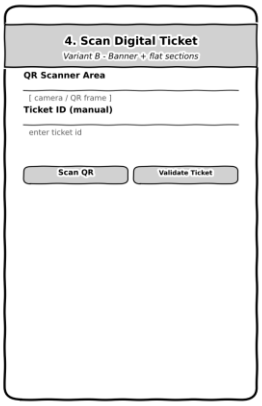
Selected Wireframe

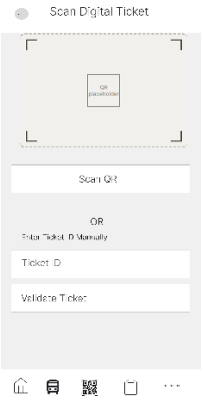


3.1.23 Scan Digital Ticket

Variant 1(selected) Variant 2 Selected Wireframe

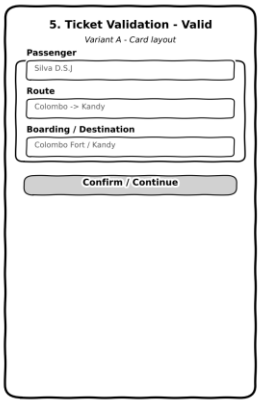






3.1.24 Ticket Validation-Valid

Variant 1(selected) Variant 2 Selected Wireframe

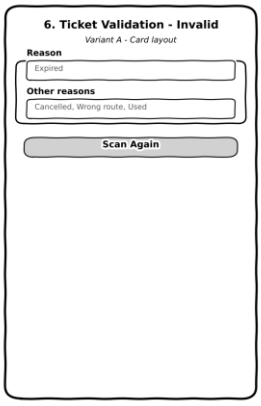




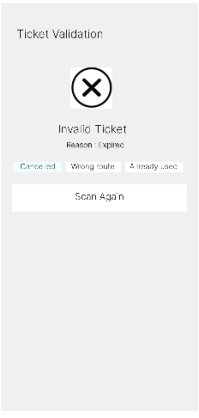


3.1.25 Ticket Validation-Invalid

Variant 1(selected) Variant 2 Selected Wireframe

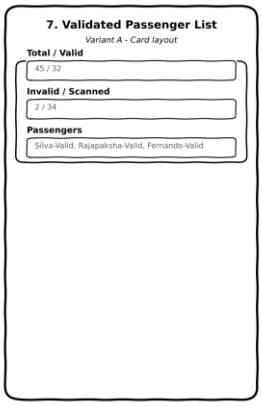


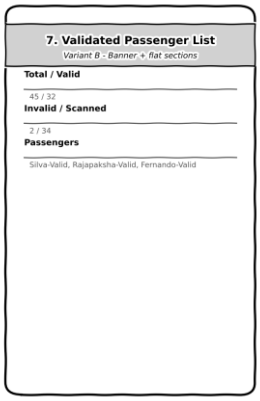


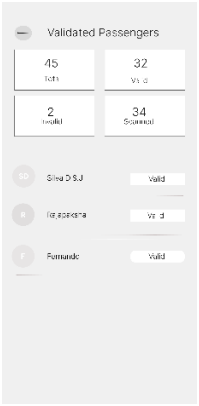


3.1.26 Validated Passenger List

Variant 1(selected) Variant 2 Selected Wireframe

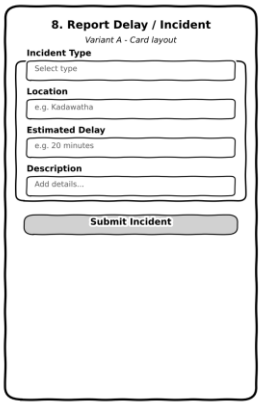




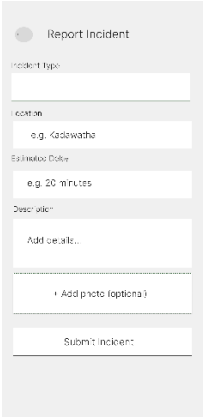


3.1.27 Report Delay / Incident

Variant 1(selected) Variant 2 Selected Wireframe



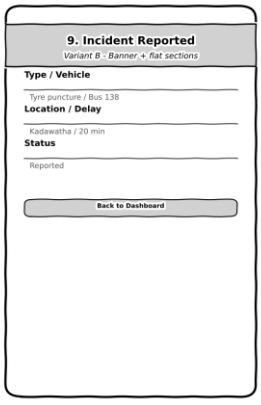


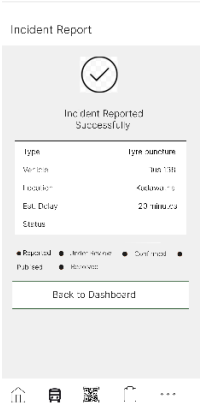


3.1.28 Incident Reported

Variant 1(selected) Variant 2 Selected Wireframe

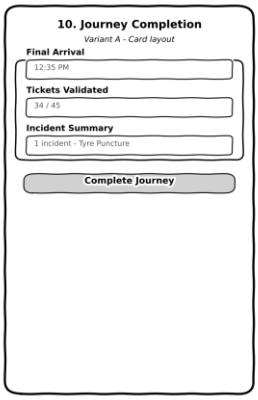




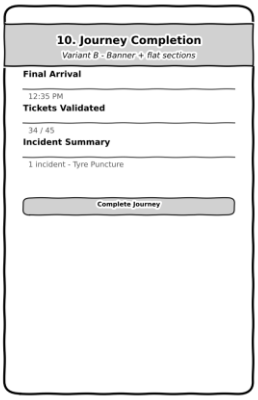


3.1.29 Journey Completion

Variant 1(selected)



Variant 2

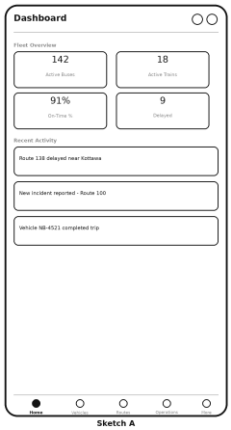


Selected Wireframe

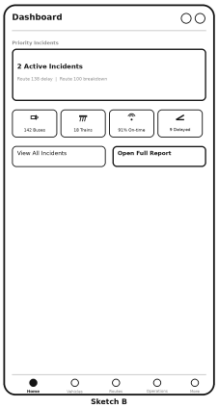


3.1.30 Transport Authority Dashboard

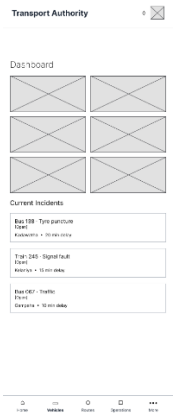
Variant 1(selected)



Variant 2



Selected Wireframe



3.1.31 Vehicle Management

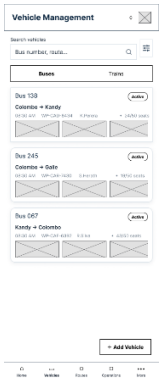
Variant 1(selected)



Variant 2

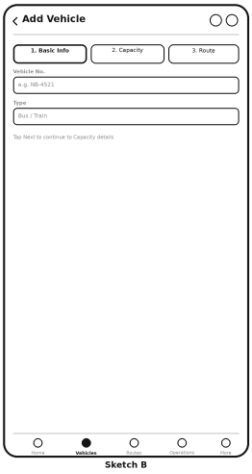


Selected Wireframe

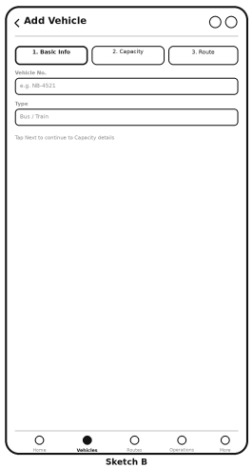


3.1.32 Add Vehicle

Variant 1(selected)



Variant 2

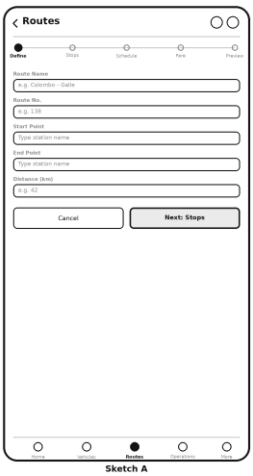


Selected Wireframe

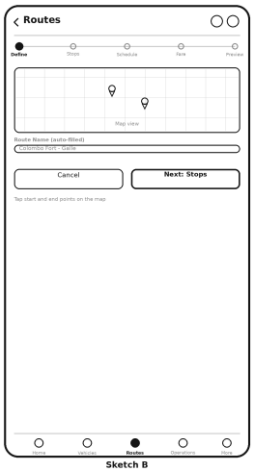


3.1.33 Routes – Define Routes

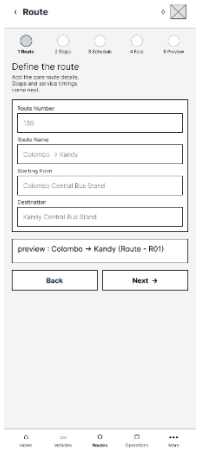
Variant 1(selected)



Variant 2

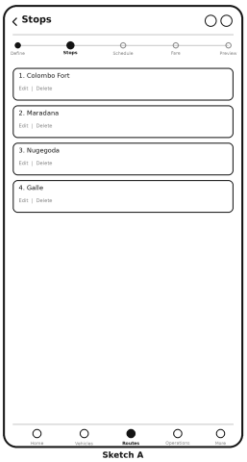


Selected Wireframe

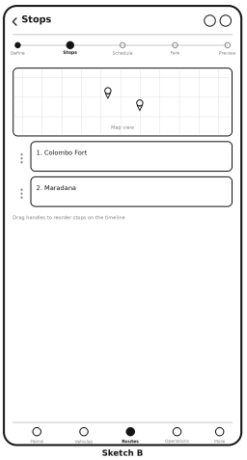


3.1.34 Routes – Stop

Variant 1(selected)



Variant 2

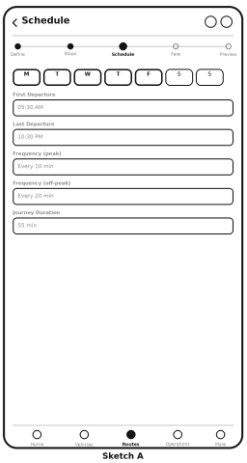


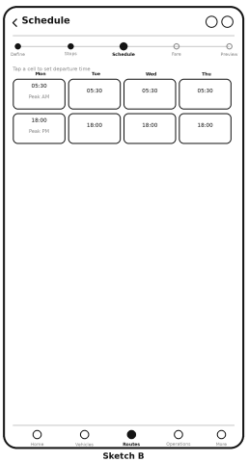
Selected Wireframe

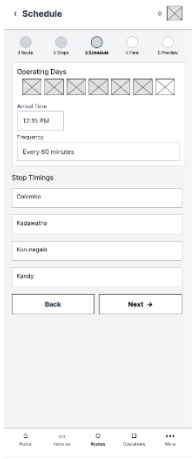


3.1.35 Routes – Schedule

Variant 1(selected) Variant 2 Selected Wireframe

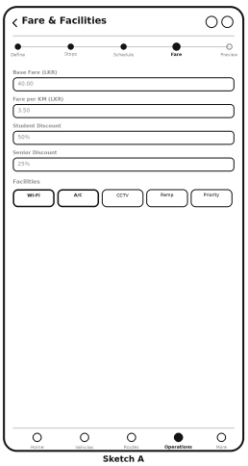


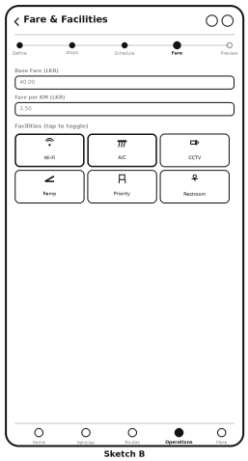


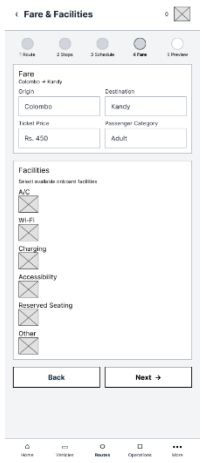


3.1.36 Fare & Facilities

Variant 1(selected) Variant 2 Selected Wireframe

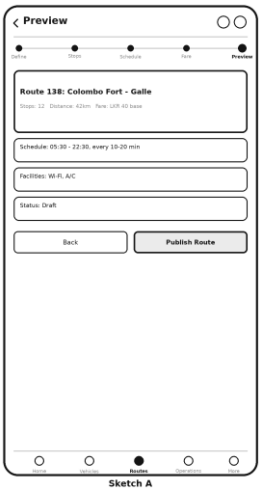




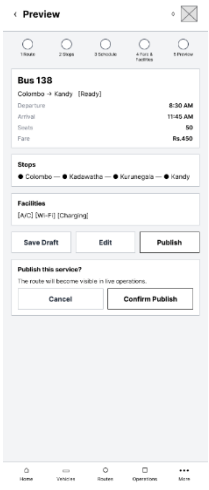


3.1.37 Routes – Preview &Publish

Variant 1(selected) Variant 2 Selected Wireframe

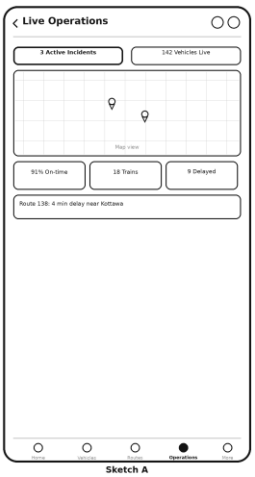




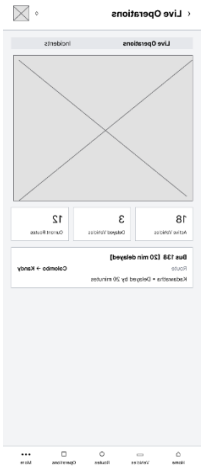


3.1.38 Live Operations

Variant 1(selected) Variant 2 Selected Wireframe

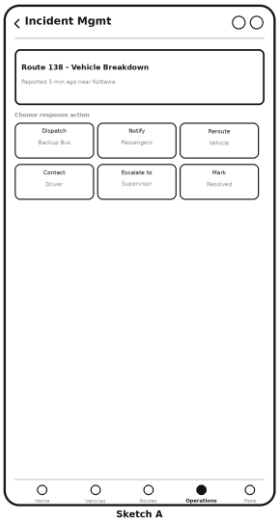




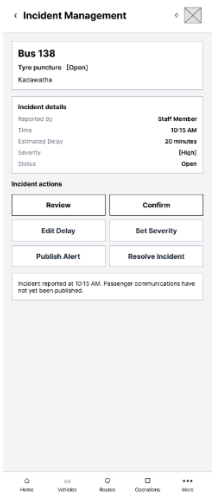


3.1.39 Incident Management

Variant 1(selected) Variant 2 Selected Wireframe

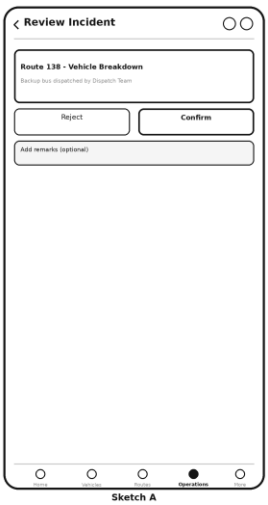


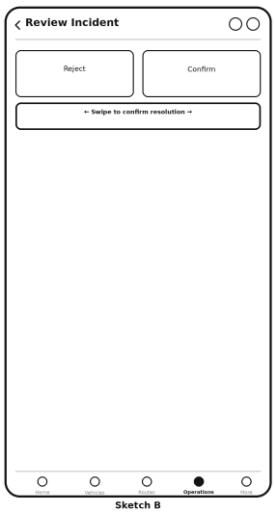


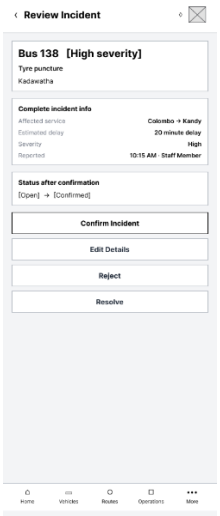


3.1.40 Review Incident

Variant 1(selected) Variant 2 Selected Wireframe







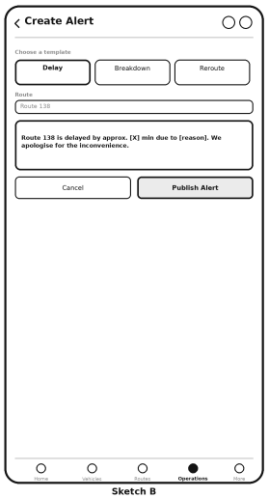
3.1.41 Create Passenger Alert

Variant 1(selected)



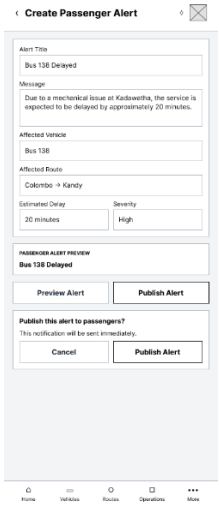
Sketch A

Variant 2



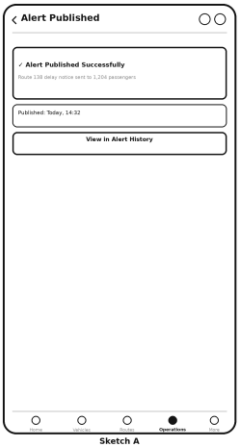
Sketch B

Selected Wireframe



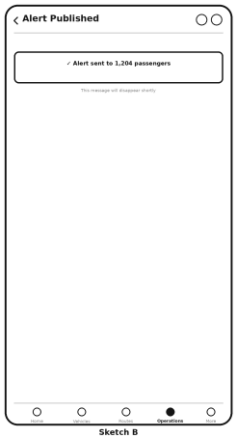
3.1.42 Alert Published

Variant 1(selected)



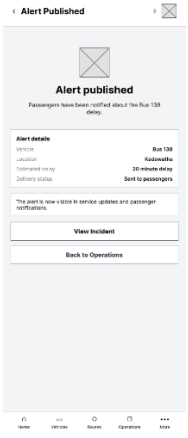
Sketch A

Variant 2



Sketch B

Selected Wireframe



3.3 Low-Fidelity Interaction Flow

Member 1 – Passenger search:

Home → Select Transport → Search → Results and Filter → Transport Details → Route and Stops or Live Tracking

Member 2 – Passenger booking:

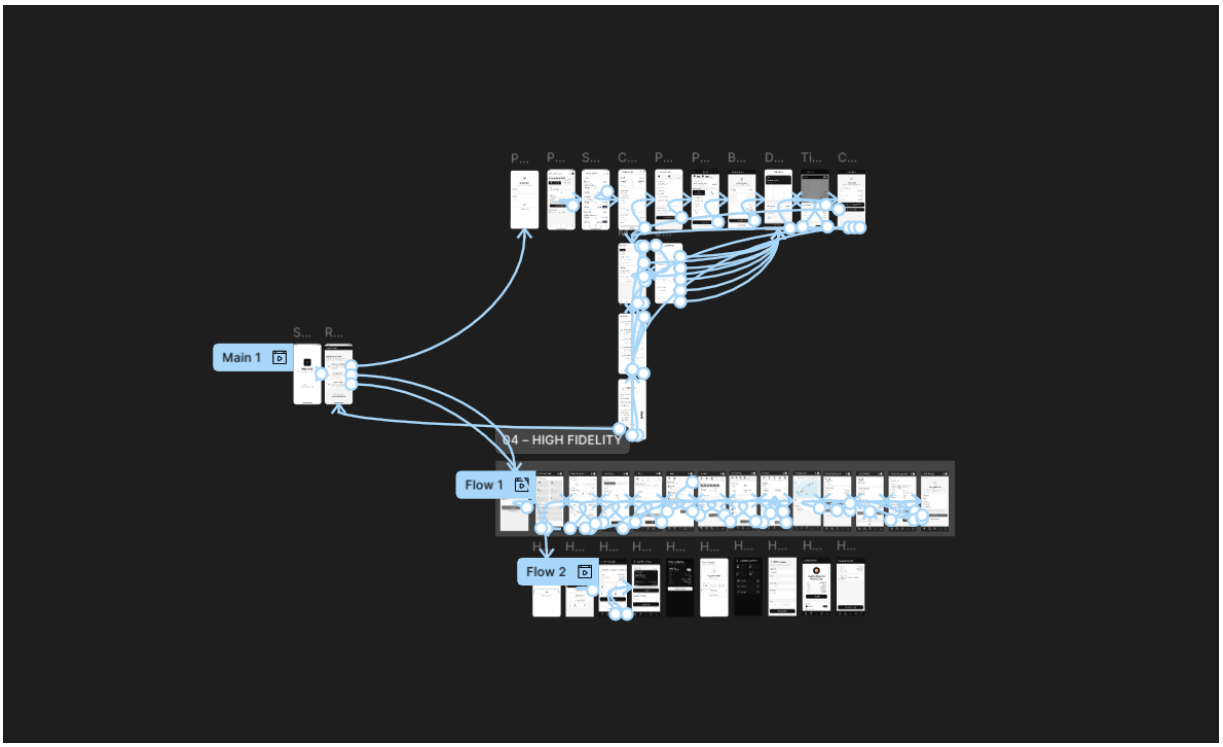
Transport Details → Journey Confirmation → Passenger Details → Payment → Booking Confirmation → Digital Ticket → Ticket Details or Cancellation

Member 3 – Conductor or Inspector:

Staff Login → Dashboard → Journey Details → Scan Ticket → Valid or Invalid Result → Passenger List → Report Incident → Complete Journey

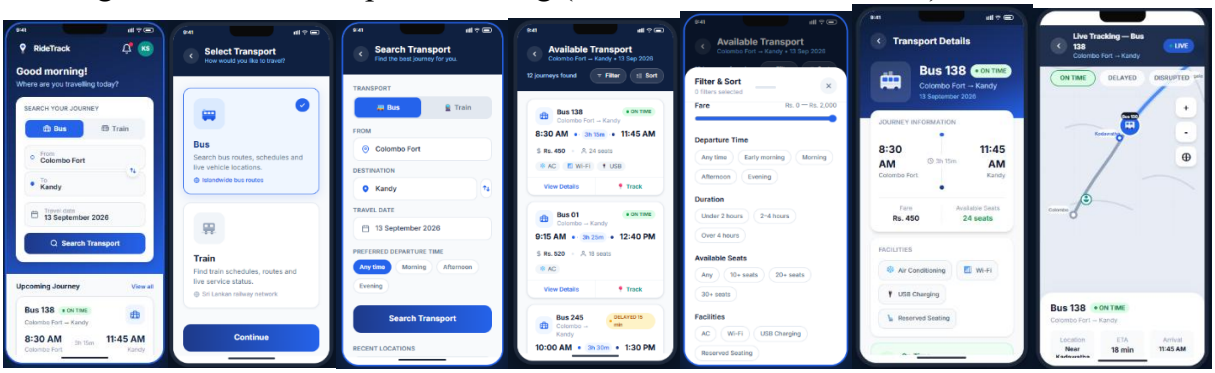
Member 4 – Transport Authority:
Dashboard → Live Operations → Vehicle, Route, Stop and Schedule Management → Incident Review → Publish Passenger Alert

3.4 Low-fidelity prototype

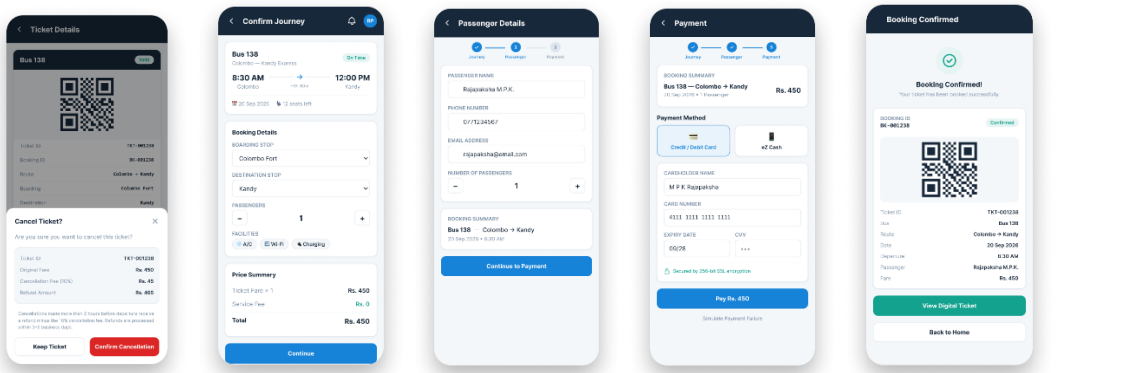


4. High-fidelity Prototype & Visual Design System and Consistency

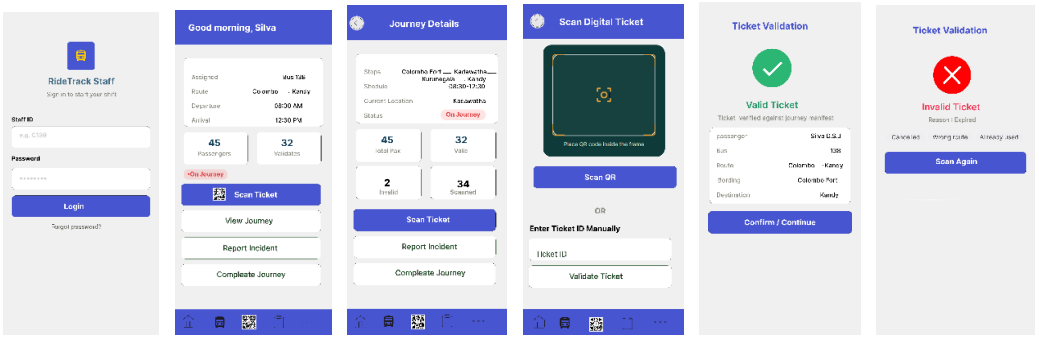
Passenger Search & Transport Tracking (Silva D S J - IT23652200)

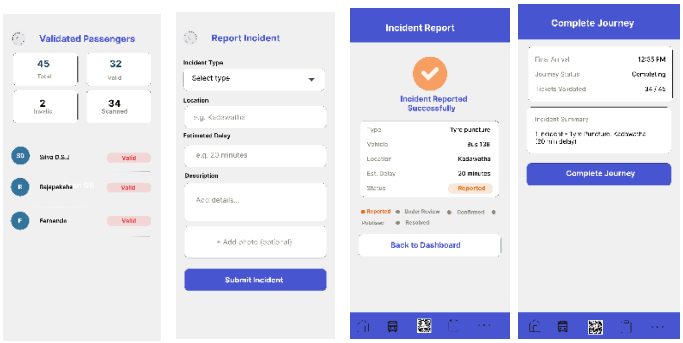


Passenger Booking & Ticket Management (Rajapaksha M.P.K - IT23665620)

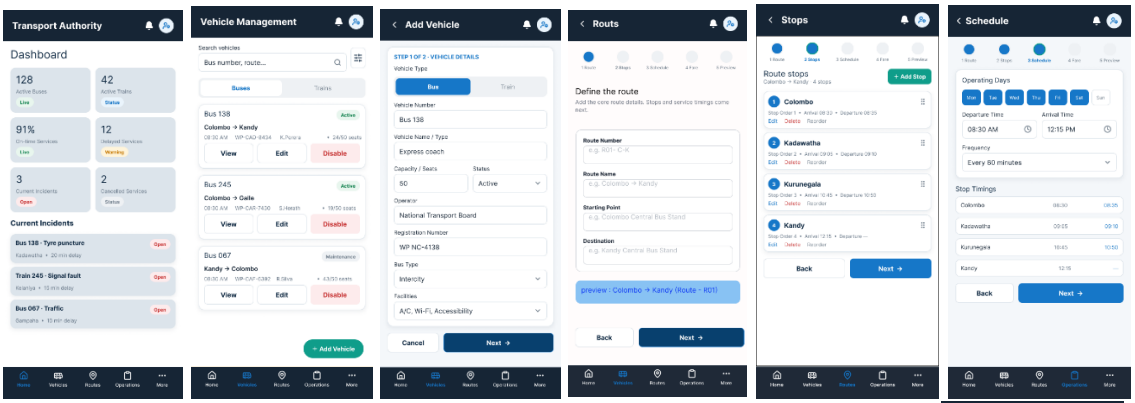


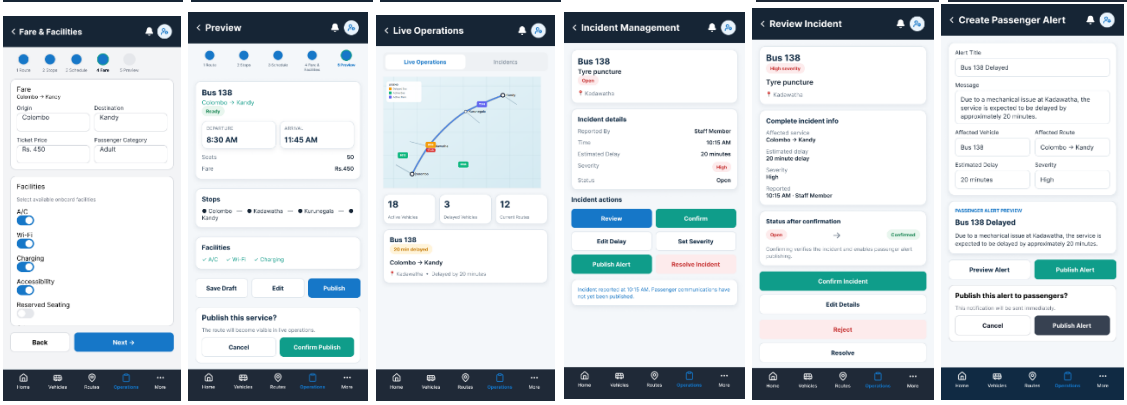
Conductor/Inspector & Ticket Validation(Herath H.M.S.G - IT23634626)

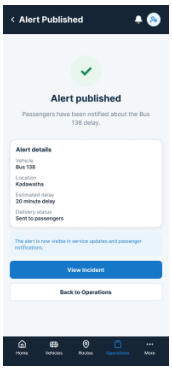




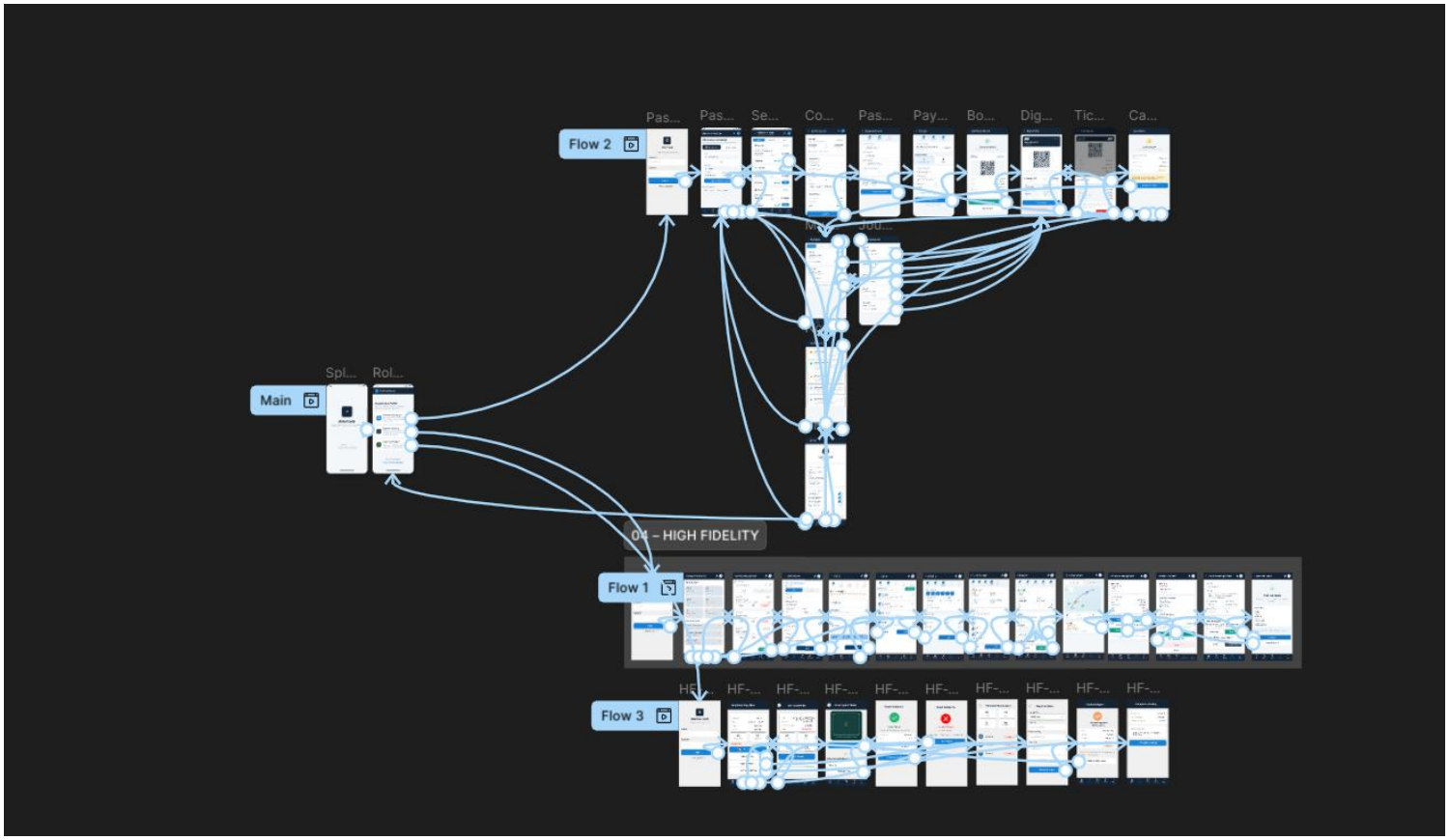
Transport Authority Management & Monitoring(ernando H L R D - IT23635302)







4.2 High-fidelity Prototype UI



5.Testing Plan and User feedback

Test Case	Test Scenario	Main Steps	Expected outcome	Success Criteria	Metrics Captured	Status (P/F)
T1 – Search & Live Tracking	Find a bus or train from Colombo Fort to Kandy and open live tracking to check its current location and ETA.	Passenger Home → Select Transport → Search Transport → Search Results & Filter → Transport Details → Live Vehicle Tracking	The correct service is located and the live map opens showing current location and ETA.	The user selects the correct service and opens live tracking without assistance, and correctly reads the ETA and on-time/delayed status.	Task completion (Yes/No); time taken; number of taps/errors; whether ETA and status were correctly interpreted.	P

T2 – Book Ticket & Digital Ticket	Book one ticket for the selected service, enter passenger details, complete payment and view the QR ticket.	Transport Details → Journey Confirmation → Passenger Details → Payment → Booking Confirmation → Digital Ticket	The booking is confirmed and the digital QR ticket is displayed with the correct details.	Booking is completed in two attempts or fewer, with no incorrect passenger or payment information, and the correct ticket is shown.	Completion rate; time taken; form validation errors; number of correction steps; confidence rating.	P
T3 – Find & Cancel Ticket	Locate a previously booked ticket in My Tickets and cancel it, understanding the refund details shown.	Home → My Tickets → Ticket Details → Cancel Ticket → Confirm Cancellation	The correct ticket is located and cancelled, and the refund amount/status is understood.	The user locates the correct ticket and completes cancellation without guidance, correctly interpreting the refund information.	Completion rate; time taken; number of backtracks; whether refund details were understood (Yes/No).	P
T4 – Scan & Validate Ticket	Log in as conductor, scan a passenger's QR ticket and interpret whether it is valid or invalid.	Staff Login → Conductor Dashboard → Journey Details → Scan Ticket → Valid/Invalid Result → Validated Passenger List	The scan returns an unmistakable valid or invalid state and the passenger list updates accordingly.	The user correctly interprets the validation result and the passenger list reflects it without assistance.	Completion rate; time taken; scan-failure fallback used (Yes/No); interpretation errors.	F
T5 – Report Incident	Report a delay incident for the current journey, providing type, location, estimated delay and description.	Conductor Dashboard → Report Incident → Enter Incident Type / Location / Delay / Description → Submit Incident	Required incident details are submitted and a confirmation screen is shown.	All required fields are completed correctly on the first attempt and the confirmation is understood.	Completion rate; time taken; number of missed/omitted fields; whether confirmation was understood.	P
T6 – Review Incident & Publish Alert	Review a reported incident and publish a passenger alert for the affected service.	Authority Dashboard → Incident Management → Review Incident → Confirm Incident → Create Passenger Alert	The incident is reviewed and an alert is published to the correct affected service.	The correct incident and affected service are identified, and the alert is published without unintended changes.	Completion rate; time taken; number of errors; whether Preview was opened before publishing (Yes/No).	F

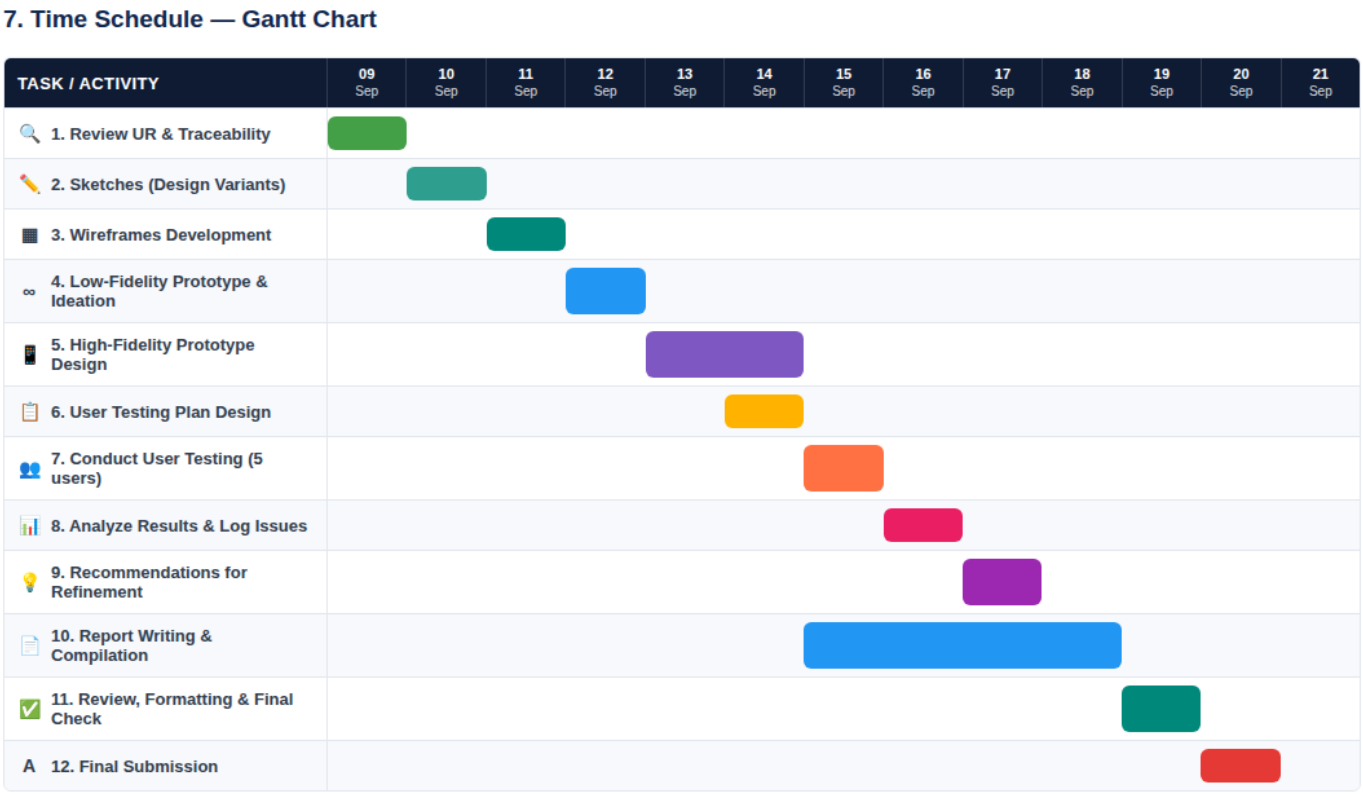
		→ Preview → Publish Alert				
T7 – Update Operational Info	Add a new vehicle to the fleet and update the schedule and fare details for a route.	Vehicle Management → Add Vehicle → Enter Details → Save → Schedule / Fare & Facilities → Edit → Save	Vehicle, schedule and fare data are saved correctly and reflected on the dashboard.	Data is saved with all required fields completed and validated in two attempts or fewer.	Completion rate; number of validation errors; time taken; satisfaction rating.	P

6.Recommendations for further refinement

Issue ID	Related Test Case	Description	Severity	Frequency	Recommendation
UI-01	T1	Users were unsure whether the "Live" status referred to real-time GPS tracking or a scheduled estimate, causing hesitation before trusting the ETA.	Medium	2 out of 5	Add a small pulsing "Live" indicator with a "last updated X sec ago" label next to the ETA.

UI-02	T2	The passenger-count stepper was easy to miss, so participants proceeded with the default of one passenger and had to go back after reaching payment.	High	3 out of 5	Move the passenger-count field above the fare summary and highlight it if left at the default before continuing.
UI-03	T3	Users expected a direct "Cancel Ticket" action on the Ticket Details screen but instead had to open a secondary confirmation view to find it.	Medium	2 out of 5	Keep a persistent "Cancel Ticket" button directly on the Ticket Details screen for upcoming journeys.
UI-04	T4	On the first scan attempt, poor lighting/angle caused scan failures for several participants, and the manual ticket-ID entry option was not immediately visible.	High	3 out of 5	Surface "Enter Ticket ID Manually" as a visible option beneath the scan frame rather than only after a failed scan.
UI-05	T5	Participants were unsure whether the incident had actually reached the Transport Authority, since the confirmation screen did not indicate visibility to authority users.	Medium	2 out of 5	Add a line on the confirmation screen such as "Visible to Transport Authority now" with a status badge.
UI-06	T6	Some participants clicked "Publish Alert" without opening the Preview step first, risking an alert being sent with incorrect details.	High	2 out of 5	Require the Preview screen to be opened at least once before the "Publish Alert" button becomes active.
UI-07	T7	When adding a new vehicle, required fields (capacity, registration number) were not visually distinguished from optional fields, leading to incomplete submissions.	Low	1 out of 5	Mark required fields with an asterisk and disable "Next" until all required fields are completed.

7. Time Schedule — Gantt Chart



Illustrative 13-day schedule matching the activities in the report's Section 7 Time Schedule table.

8. References

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9. Appendix

5.3 Visual Design System and Consistency

● Low Fidelity Prototype Video →
https://www.figma.com/design/TgS8Rf3MSdO00Z26poSm9V/RideTrack?node-id=11-2&t=BLeX1GujesW9atF5-1

● High Fidelity Prototype Video →

https://www.figma.com/design/TgS8Rf3MSdO00Z26poSm9V/RideTrack?node-id=0-1&t=BLeX1GujesW9atF5-1

User Testing Recordings Folder

https://drive.google.com/drive/folders/1OiAK4fKyYjbMMLBRIdk1AB3rGyQDLVfk?usp=sharing